Signals and Systems Lecture-4: Energy and Power

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Energy of continuous-time signals (1)

The energy of a continuous-time signal $s(t)$ is defined as the area under the square of the magnitude of the signal:

$$E = \int_{-\infty}^{+\infty} |s(t)|^2 \, \mathrm{d}t$$

- If $s(t)$ stands for a voltage, then E corresponds to the electrical energy dissipated when $s(t)$ is applied across a resistance of 1 Ohms.
  - In other words, the actual physical energy delivered by the signal $s(t)$ is defined to be **proportional** (but not necessarily equal) to E.
- The energy of some signals can be infinite.
- A signal that has a finite energy is called an **energy signal**.

Energy of continuous-time signals (2)

Example: Calculate the energy of the unit rectangular function and the unit step function.

- The energy of the rect function is given by:

$$E = \int_{-\infty}^{+\infty} \mathrm{rect}^2(t)\mathrm{d}t$$
$$= \int_{-\frac{1}{2}}^{+\frac{1}{2}} (1)^2 \mathrm{d}t = 1$$

  - Consequently, the rect signal is an energy signal.
- The energy of the step function is given by:

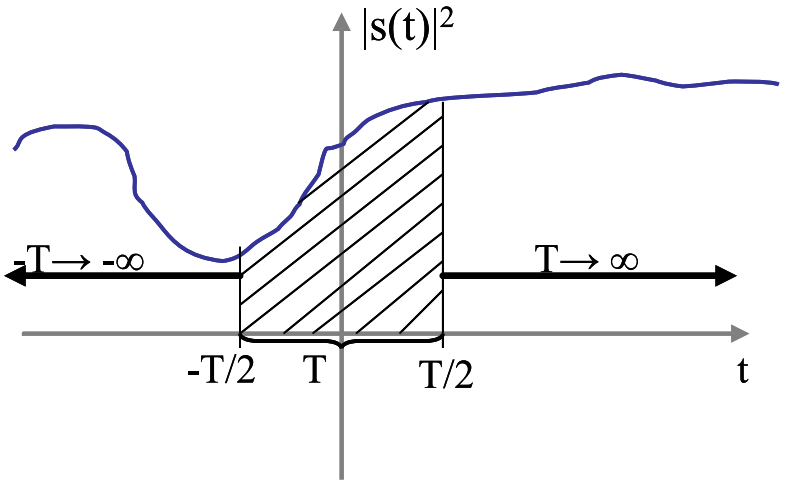
$$E = \int_{-\infty}^{+\infty} u^2(t)\mathrm{d}t$$
$$= \int_{0}^{+\infty} (1)^2 \mathrm{d}t = +\infty$$

  - Consequently, the step signal is not an energy signal.

Average Power of continuous-time signals (1)

- For signals that have an infinite energy, it is more convenient to deal with the average power of the signal.
- The average power of a continuous-time signal $s(t)$ is given by:

$$P = \lim_{T \to +\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{+\frac{T}{2}} |s(t)|^2 \, dt$$



- A signal that has an infinite energy but a finite average power is called a **power signal**.

Average Power of continuous-time signals (2)

Example: Calculate the average power of the unit step function.

- The average power of $u(t)$ is given by:

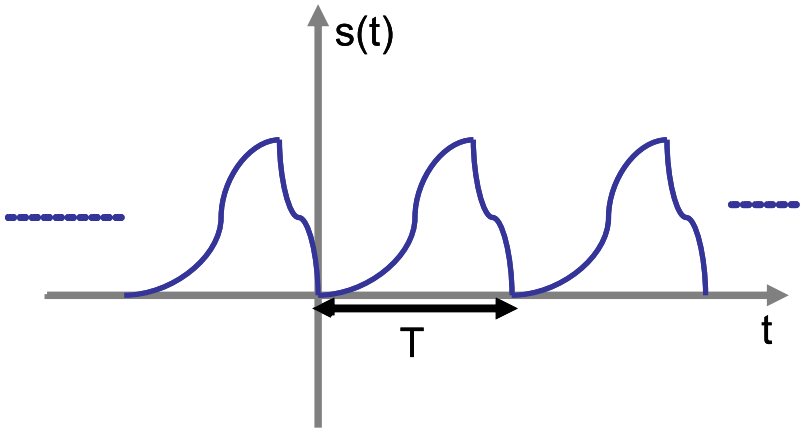
$$
\begin{aligned}
P &= \lim_{T \to +\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{+\frac{T}{2}} u^2(t) \mathrm{d}t \\
&= \lim_{T \to +\infty} \frac{1}{T} \int_{0}^{+\frac{T}{2}} 1.\mathrm{d}t \\
&= \lim_{T \to +\infty} \frac{1}{T} \frac{T}{2} = \frac{1}{2}
\end{aligned}
$$

  - $u(t)$ has an infinite energy and a finite average power; consequently, it is a power signal.

Average Power of periodic continuous-time signals

- A CT signal is periodic with a period T if:

$$s(t) = s(t + nT) \;\; \forall \;\; n \in \mathbb{Z}$$



- For periodic signals, the expression of the average power can be simplified:

$$P = \frac{1}{T}\int_{T} |s(t)|^2 \mathrm{d}t = \frac{1}{T}\int_{t_0}^{t_0+T} |s(t)|^2 \mathrm{d}t \quad \forall \quad t_0$$

  - $\int_T$ means that the integral can be calculated over any interval having a duration T.

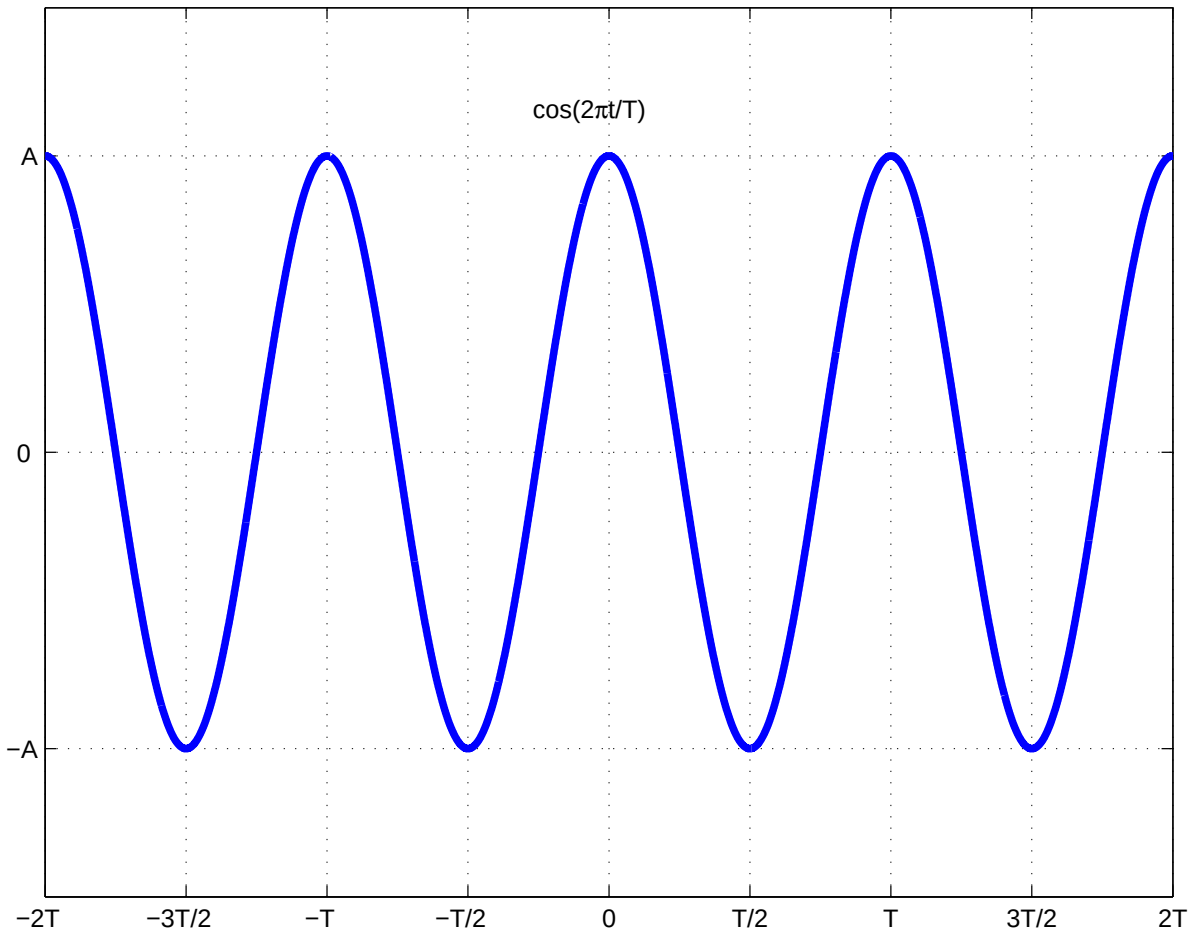
The cosine function (1)

$$f(t) = A\cos(2\pi ft + \theta)$$

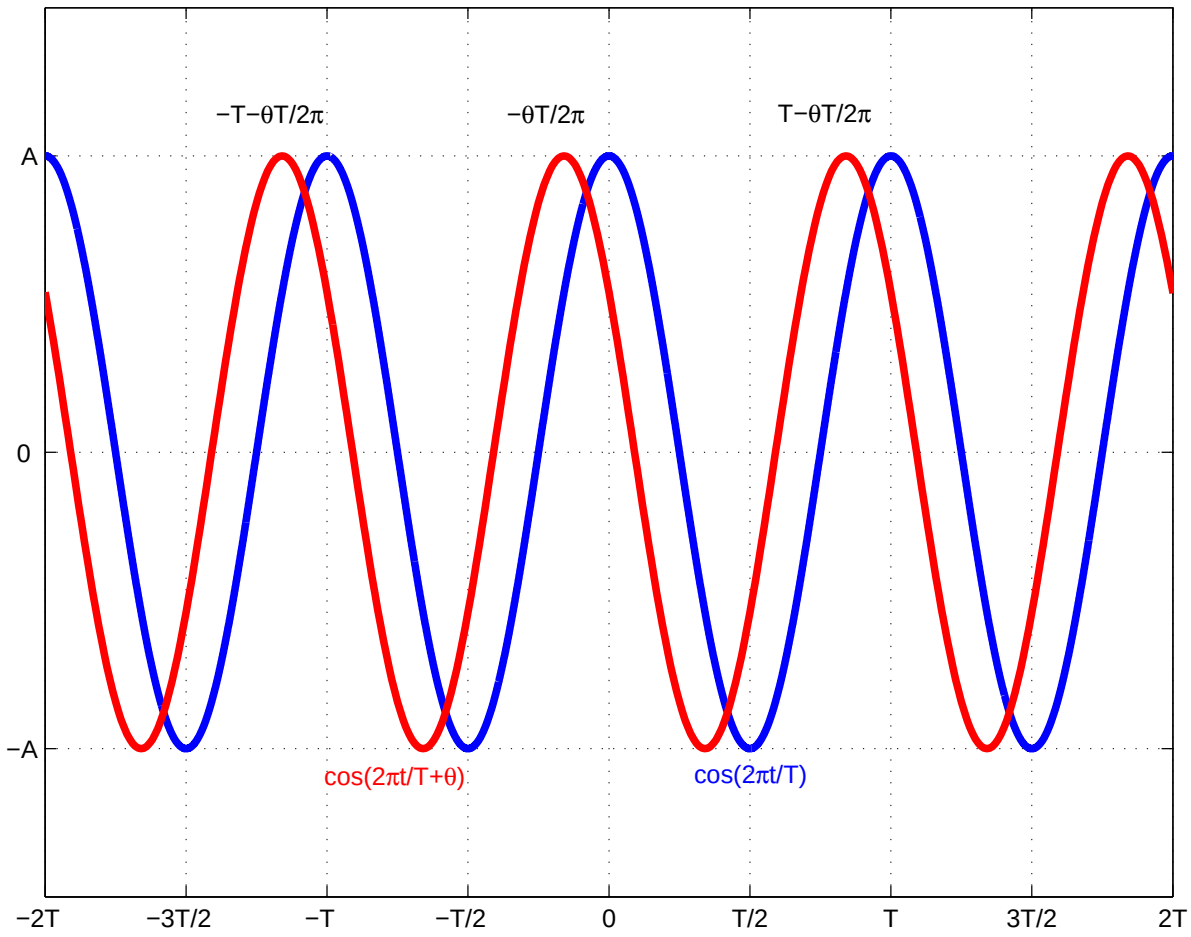
- where:
  - A is the maximum amplitude of the signal ($-A \leq f(t) \leq A \;\; \forall \;\; t$)
  - f is the frequency of the sinusoidal function (in Hz).
  - θ is the phase of the signal (in radians).
- $f(t)$ is a periodic signal with a period: $T = \frac{1}{f}$.
  - $f(t) = 0$ for $2\pi ft + \theta = \frac{\pi}{2} + k\pi \;\Rightarrow\; t = \frac{2k+1}{4f} - \frac{\theta}{2\pi f}$
  - $f(t) = A$ for $2\pi ft + \theta = 2k\pi \;\Rightarrow\; t = \frac{k}{f} - \frac{\theta}{2\pi f}$
  - $f(t) = -A$ for $2\pi ft + \theta = \pi + 2k\pi \;\Rightarrow\; t = \frac{2k+1}{2f} - \frac{\theta}{2\pi f}$

  for all integer values of k.

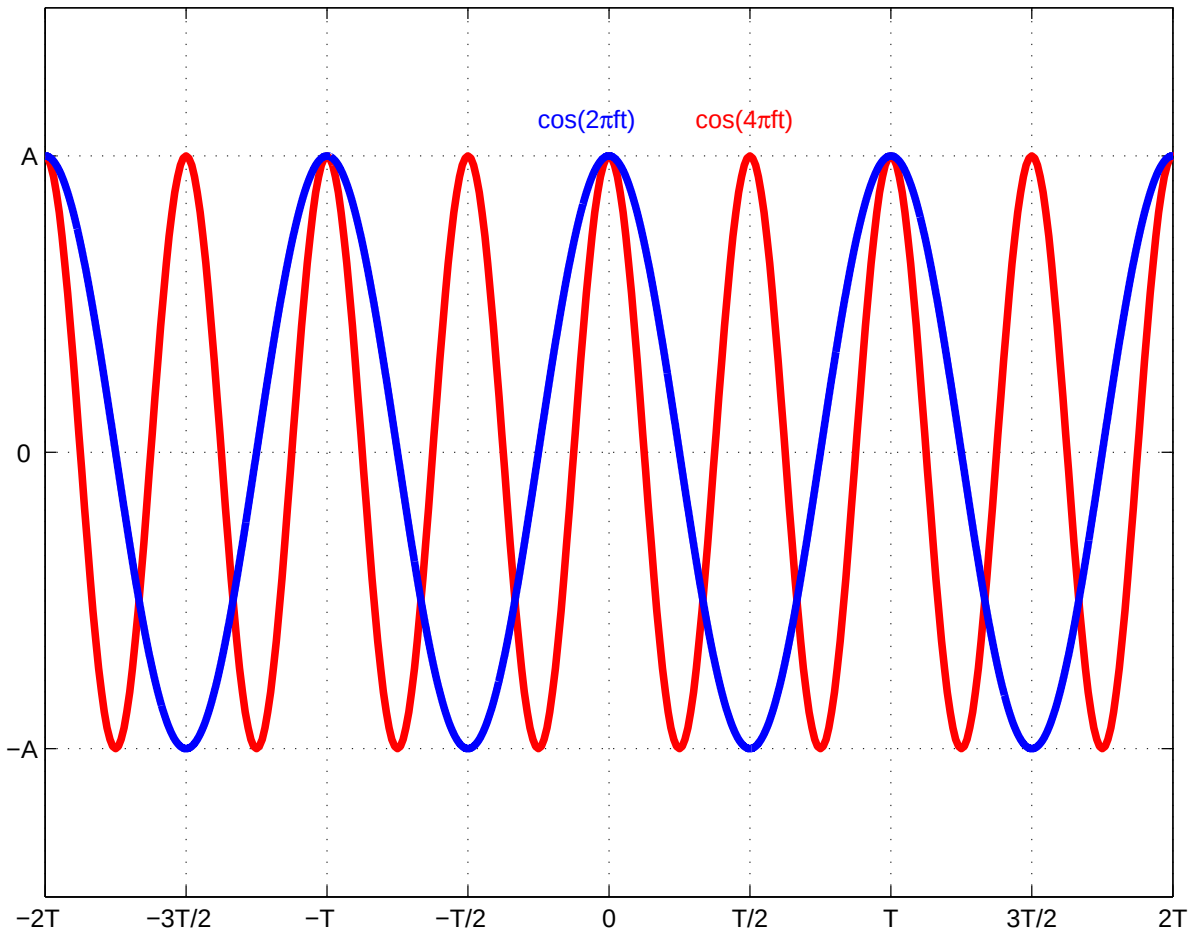
The cosine function (2)



The cosine function (3)



The cosine function (4)



Energy of the sinusoidal function

- The energy of $f(t)$ is given by:

$$E = \int_{-\infty}^{+\infty} f^2(t)\mathrm{d}t = \int_{-\infty}^{+\infty} A^2 \cos^2(2\pi f t + \theta)\mathrm{d}t$$

- Given that: $\cos(2\theta) = 2\cos^2(\theta) - 1 \Rightarrow \cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$
- Then:

$$E = \frac{A^2}{2}\left\{\underbrace{\int_{-\infty}^{+\infty} \mathrm{d}t}_{+\infty} + \underbrace{\int_{-\infty}^{+\infty} \cos(4\pi f t + 2\theta)\mathrm{d}t}_{\text{finite}}\right\} = +\infty$$

Power of the sinusoidal function (1)

- Given that the sinusoidal function is periodic:

$$
\begin{aligned}
P &= \frac{1}{T}\int_T f^2(t)\mathrm{d}t = \frac{1}{T}\int_T A^2\cos^2(2\pi ft+\theta)\mathrm{d}t \\
&= \frac{A^2}{2T}\left[\int_{t_0}^{t_0+T}\mathrm{d}t + \int_{t_0}^{t_0+T}\cos(4\pi ft+2\theta)\mathrm{d}t\right] \\
&= \frac{A^2}{2T}\left[[t]_{t_0}^{t_0+T} + \left[\frac{1}{4\pi f}\sin(4\pi ft+2\theta)\right]_{t_0}^{t_0+T}\right] \\
&= \frac{A^2}{2T}\left[T + \frac{1}{4\pi f}[\sin(4\pi ft_0 + \underbrace{4\pi fT}_{=4\pi} + 2\theta) - \sin(4\pi ft_0+2\theta)]\right] \\
&= \frac{A^2}{2T}\left[T + \frac{1}{4\pi f}[\sin(4\pi ft_0+2\theta) - \sin(4\pi ft_0+2\theta)]\right] = \frac{A^2}{2}
\end{aligned}
$$

- $4\pi fT = 4\pi$ since $T = 1/f$.
- $\sin(\theta + 2k\pi) = \sin(\theta)$.

Power of the sinusoidal function (2)

- Consequently, the sinusoidal signal is a power signal.
- The power of the sinusoidal signal depends only on the amplitude A (not on the frequency f and the phase θ).

Energy and Power of the ramp function

Example: Find the energy and power of the unit ramp function.

- The energy is given by:

$$E = \int_{-\infty}^{+\infty} \text{ramp}^2(t)\text{d}t = \int_{0}^{+\infty} t^2 \text{d}t = \left[\frac{t^3}{3}\right]_0^{+\infty} = \infty$$

implying that the ramp function is not an energy signal.

- The power is given by:

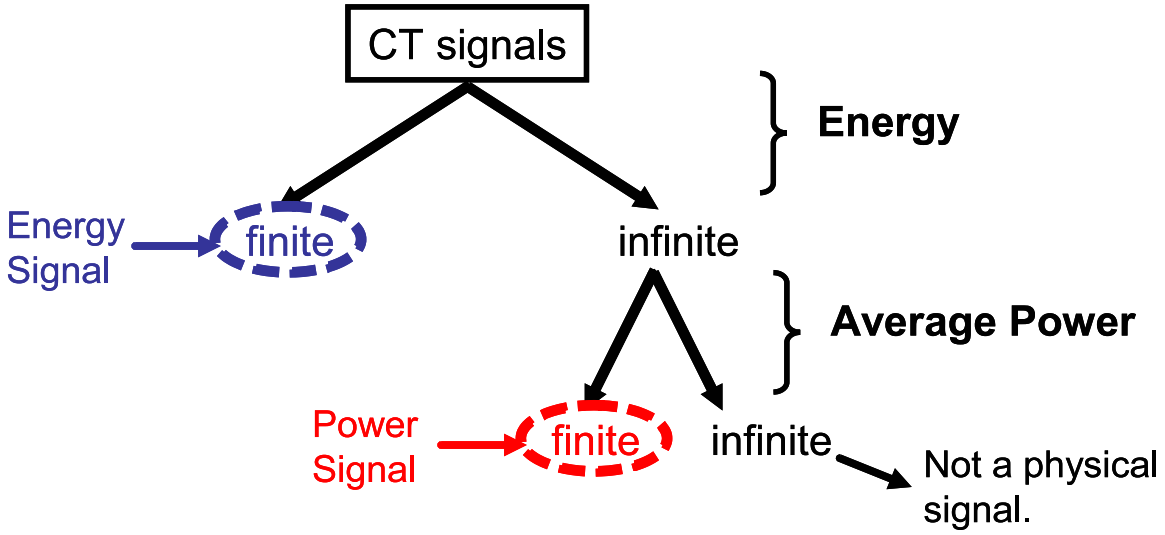
$$P = \lim_{T\to+\infty} \frac{1}{T} \int_{-T/2}^{T/2} \text{ramp}^2(t)\text{d}t = \lim_{T\to+\infty} \frac{1}{T} \int_{0}^{T/2} t^2 \text{d}t$$

$$= \lim_{T\to+\infty} \frac{1}{T} \left[\frac{t^3}{3}\right]_0^{T/2} = \lim_{T\to+\infty} \frac{1}{T} \left[\frac{T^3}{24}\right] = \lim_{T\to+\infty} \frac{T^2}{24} = +\infty$$

implying that the ramp function is not a power signal.

Classification of continuous-time signals

Classification of CT signals:



- Energy signals: $\mathrm{rect}(t)$, $\mathrm{tri}(t)$, ...
  - Note that these signals have a zero average power.
- Power signals: $u(t)$, sinusoidal functions, ...
- Signals having infinite energy and infinite average power: $\mathrm{ramp}(t)$, e^t, ...